STANDARD BANK

Multi-Branch Banking Management System

User Guide/README.txt

C++ | 2026

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1. System Overview

The Standard Bank Multi-Branch Banking Management System is a C++ application designed to simulate core banking operation across multiple branches. It is still a prototype but it supports two staff roles – Branch Managers and Tellers – and provide a full customer account management, transaction processing and reporting capabilities.

Key Features:

- 3 branch location: Sandton, Bloemfontein and Constantia
- Role-based access: Branch Managers and Tellers
- 10 intern's tellers + 3 branch managers
- 4 account type: Savings, Cheque, Fixed Deposit and Student
- Encrypted PINs and passwords stored in Binary Files
- Full transaction support: Deposit, Withdrawal, Fund Transfer
- Comprehensive reporting: Daily, Summary and Performance reports
- Data persistence across session with automatic backup
- CSV and text export functionality

2. System Files Explained

The systems automatically creates and manages the following files in the same folder:

- Tellers.dat | Binary | Stores encrypted teller and manager credentials
- Customers.dat | Binary | Stores all customer accounts with encrypted PINs
- Transactions.dat | Binary | Logs every transaction performed
- Branches.dat | Binary | Branch performance data
- System_Config.txt | Text | System settings, interest rates and last login info
- Customers_backup.dat | Binary | Automatic backup of customers.dat
- Customer_export.csv | CSV | Customer data exported in a spreadsheet
- Customer_export.txt | Text | Customer data exported in a readable format
- Daily_transaction_report.txt | Text | Appended daily transaction report per branch

3. First Run – What to Expect

When you run the application for the very first time, the system will automatically set itself up:

```
============================================== STANDARD BANK MULTI-BRANCH
SYSTEM v1.0 ============================================== First run: seeding
staff credentials... Seeded: 3 managers + 10 tellers. No customer data. Starting
fresh. Please select your role: 1. Branch Manager 2. Teller 3. Exit System
Choice:
```

The System automatically:

- ✓ Creates tellers.dat with 3 managers and 10 tellers
- ✓ Creates system_config.txt with default settings
- ✓ Initialises the 3 branch network (Sandton, Bloemfontein, Constantia)

4. Default Login Credentials

All default credentials are seeded automatically on the first run. Use these to log in during testing and demonstration.

Branch Managers

Staff ID	Password	Branch	Name
M001	Manager001!	Sandton (018105)	Sipho Dlamini
M002	Manager002!	Bloemfontein (055034)	Priya Naidoo
M003	Manager003!	Constantia (025309)	Anke Van Der Berg

Tellers

Staff ID	Password	Branch	Name
T001	Teller001!	Sandton (018105)	Lerato Mokoena
T002	Teller002!	Sandton (018105)	James Sithole
T003	Teller003!	Sandton (018105)	Fatima Osman
T004	Teller004!	Sandton (018105)	Thabo Khumalo
T005	Teller005!	Bloemfontein (055034)	Rudo Mhlanga
T006	Teller006!	Bloemfontein (055034)	Nomsa Zulu
T007	Teller007!	Bloemfontein (055034)	Deon Pretorius
T008	Teller008!	Constantia (025309)	Ayasha Pillay
T009	Teller009!	Constantia (025309)	Kagiso Sithole
T010	Teller010!	Constantia (025309)	Mia Fourie

Note: Managers can reset tellers password from their menu.

5. Branch Manager Guide

Branch Managers have the highest level of access. Each manager can only manage tellers and customers within their branch.

Logging in as a Manager:

```
Please select your role: 1. Branch Manager <-- type 1 2. Teller 3. Exit System
---- Standard Bank Manager Login ---- Staff ID : M001 Password : Manager001!
Welcome, Sipho Dlamini!
```

Manager Menu Options

Option	Action	Description
1	Register New Customer	Opens the customer registration form
2	View/Search Customer	Search by account number, name, or SA ID
3	Process Transaction	Deposit, withdraw, transfer on behalf of customer
4	Add New Teller	Create a new teller account in your branch
5	Remove a Teller	Delete a teller from your branch
6	View Branch Teller	List all tellers assigned to your branch
7	Reset Teller Password	Change a teller's password
8	Branch Report	Access all reporting tools
9	Logout	Return to the role selection screen

6. Tellers Guide

Tellers handle day-to-day customer operations. They do not have access to branch reports or teller management – those are manager only functions.

Logging in as a Teller:

```
Please select your role: 1. Branch Manager 2. Teller <-- type 2 3. Exit System
---- Standard Bank Teller Login ---- Staff ID : T001 Password : Teller001!
Welcome, Lerato Mokoena!
```

Teller Menu Options

Option	Action	Description
1	Register New Customer	Opens the full customer registration form
2	View/Search Customer	Search customer by account number, name, or SA ID
3	Process Transaction	Handle deposits, withdrawals, transfers for a customer
4	Logout	Return to role selection screen

Processing a Transaction (Option 3)

- Step 1: The system asks you to search for the customer
- Step 2: Find them by account number, name, or SA ID
- Step 3: Enter the customer's 5-digit PIN for verification
- Step 4: Choose the transaction type from the submenu
- Step 5: Follow the prompts to complete the transaction
- Step 6: Choose option 7 to return to the main menu

Note: The customer must be physically present to provide their PIN. Tellers cannot process the transaction without PIN Verification.

7. Customer Account Registration

Only authenticated tellers or managers can register new customers accounts. All fields are validated before the account is created.

Registration Steps:

```
===== New Customer Registration ===== First Name : Jane Surname : Doe SA ID
Number (13 digits): 7208290123456 Contact Number (10 digits): 0821234567 Email
Address: janedoe@gmail.com Date of Birth (DD/MM/YYYY): 29/08/1972 Physical
Address: 18 Whitfield Street, Sandton Account Types: 1. Savings (Min R100,
Interest 4.25%) 2. Cheque (Min R100, No interest) 3. Fixed Deposit (Min R1000,
Interest 8.05%) 4. Student (Min R50, Interest 4.75%, Zero Fees) Choice: 1
Initial Deposit (Min R100.00): R5000 ===== Registration Complete! ===== Account
Number : ACC-018105-20272 Temporary PIN : 31116 (Hand to customer securely)
Account Type : Savings Initial Deposit: R5000.00
```

Validation Rules

Field	Rule
SA ID Number	Must be exactly 13 numeric digits
Contact Number	Must be exactly 10 numeric digits
Email Address	Must contain @ and a domain (e.g. .com)
Date of Birth	Must be in DD/MM/YYYY format
Physical Address	Must be at least 5 characters
Initial Deposit	Must meet the minimum for the chosen account type

8. Customer Login & Transactions

Customers access their accounts through a teller or manager-assisted session, or through a self-service session launched from within the transaction menu.

Teller-Assisted Login (Option 3 from Teller/Manager Menu)

- The teller searches for the customer using Option 2 first
- Then selects Option 3 — Process Transaction
- Enters the customer's account number or name to find them
- The customer verbally provides their PIN to the teller
- The teller types the PIN — access is granted if correct

Self-Service Session

From within the transaction submenu, the teller can select Option 6 — Start Customer Self-Service Session. The customer then takes over the terminal and manages their own session:

```
===== Self Service: Jane Doe ===== 1. View Balance 2. Deposit 3. Withdraw 4. Fund Transfer 5. View Statement 6. Change PIN 7. Logout Choice:
```

Fund Transfer

- Type the recipient's name (case-insensitive)
- If one match is found — it confirms the name and account, then proceeds
- If multiple matches found — all are listed and you must enter the account number to confirm
- Enter the transfer amount

Changing PIN (Self-Service Only)

- Customer enters their current PIN first for verification
- Then enters and confirms the new 5-digit PIN
- PIN must be exactly 5 digits and must be unique across all accounts

Login Attempt Limit

If a customer enters the wrong PIN 3 times in a row, the account is automatically locked. A teller or manager must unlock it by finding the customer record and selecting 'Unlock if locked? (y/n): y'.

9. Account Types & Interest Rates

Account Type	Min Deposit	Annual Interest
Savings	R100.00	4.25%
Cheque	R100.00	None
Fixed Deposit	R1,000.00	8.05%
Student	R50.00	4.75%

Fixed Deposit Early Withdrawal

If a customer withdraws from a Fixed Deposit account before maturity, a 10% penalty is automatically applied to the withdrawal amount. The system will warn the customer before proceeding:

```
WARNING: Early withdrawal penalty of 10% applies to Fixed Deposit! Amount
requested: R1000.00 Penalty (10%) : R100.00 Total deducted : R1100.00
```

10. Report Guide

Reports are accessible to Branch Managers only (Option 8 from the Manager Menu). Each manager can only view reports for their own branch.

Report	Option	Description	Output File
Daily Transaction Report	1	All transactions for today in branch	daily_transaction_report.txt
Customer Account Summary	2	Customer counts, vault total, interest forecast rates	Screen Only
Branch Performance Report	3	All branches compared by total funds	branches.dat
Inter-Branch Comparison	4	Quick view of all branch balances	Screen only
Export to CSV	5	Full customer list in CSV format	customer_export.csv
Export to Text File	6	Full customer list in text format	customer_export.txt

Running the Daily Transaction Report

```
===== Branch Reports ===== 1. Daily Transaction Report <-- type 1 ... ==========
Daily Transaction Report ========== Branch : 018105 Date : 24/04/2026
---------------------------------------------- Jane Doe | Deposit | R5000.00 |
24/04/2026 09:15 John Doe | Withdrawal | R200.00 | 24/04/2026 09:22
---------------------------------------------- Total In : R5000.00 Total Out :
R200.00 Net : R4800.00
```

11. File Exports

The system can export customer data in two formats from the Branch Reports menu.

CSV Export (customer_export.csv)

Opens in Microsoft Excel or Google Sheets. Contains:

- Name, ID Number, Email, Date of Birth
- Account Number, Account Type, Branch Code, Balance

Text File Export (customer_export.txt)

A human-readable formatted report. Contains all customer details in a structured layout, one customer per block, with a timestamp header.